

# The Value of Delayed Information: Predictive-Information Envelopes for Age-Limited Control

Exact Bayes envelope generally; mutual-information bound universally;  
closed form in the symmetric Markov case

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Working draft — v0.2 (complete draft: theory, hinge law, envelope comparison, battery,  
related work)

## Abstract

A controller that acts on information of age  $D$  — a sensing delay, a telemetry lag, a stale cache — can outperform the best static policy only to the extent that the delayed observation still *predicts* the current state. Our companion systems work [1] proves this exactly for symmetric two-state Markov capacity ( $\Delta(D) = \frac{1}{2}r(D)$ , the autocorrelation at the sensing lag) with a  $\beta$ -mixing converse. This paper supplies the general theory. We (i) isolate the exact object — the *Bayes envelope*  $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$ , the expected excess value of posterior over prior beliefs; (ii) prove the universal *predictive-information envelope*  $0 \leq \Delta_Y \leq \Omega\sqrt{\frac{1}{2}I(X;Y)}$  for any bounded objective with oscillation  $\Omega$ , by Pinsker’s inequality applied to the posterior process; (iii) derive the *bit-budget corollary*: a telemetry path delivering  $b$  bits per decision caps feedback value at  $\Omega\sqrt{(b \ln 2)/2}$  regardless of what is encoded — compression can preserve but never create feedback value; (iv) show the envelope is *tight to leading order* in the canonical case: for the symmetric binary chain  $I = \frac{r^2}{2} + O(r^4)$ , so the bound reads  $\Omega(\frac{r}{2} + O(r^3))$  against the exact  $\frac{1}{2}r$ ; (v) prove by a two-bit construction that no universal equality  $\Delta = f(I)$  can hold: mutual information is a scalar, decision value is a geometry, and two observations with identical  $I$  can carry all or none of the decision-relevant distinction; and (vi) give a *complete* exactness theory for binary state — for any bounded objective the value of a delayed sample is a *hinge law* in the lag covariance, linear beyond a dead zone and identically zero inside it, with the companion’s law as the symmetric zero-threshold case. A nine-series battery (acoustics, bioacoustics, space weather, seismology, plus synthetic controls) confirms the law where lag-1-matched surrogates say it must hold, observes the envelope’s predicted  $O(r^2)$  tightness on real data, and inverts the theorem into a certified lower bound on history information where the processes are non-Markov.

## 1 Introduction

Everywhere a controller acts on stale state — a scheduler admitting jobs against capacity sensed seconds ago, a trader on delayed quotes, a relay on an outdated channel estimate — the same question recurs: *what is the value of information of age  $D$ ?* The age-of-information literature [7] measures freshness; freshness, however, has no decision semantics: a maximally fresh sample of an i.i.d. process is worthless, and a stale sample of a slowly-mixing one is nearly as good as

the present. Our companion systems paper [1] made this precise in one setting, proving that for symmetric two-state Markov capacity the advantage of age- $D$  feedback over the best static policy is *exactly*  $\Delta(D) = \frac{1}{2}r(D)$  — the autocorrelation at the sensing lag — and measuring the law on a live GPU cluster. This paper supplies the theory that result belongs to. The thesis is one line:

*Age matters only through predictive information, and predictive information matters only through the Bayes envelope.*

Concretely: **(C1)** we isolate the exact object — the Bayes envelope  $\Delta_Y = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu})$ , classical in decision theory [2, 4, 3] — and prove the universal *predictive-information envelope*  $\Delta_Y \leq \Omega\sqrt{I(X;Y)/2}$  (Theorem 2), with the *bit-budget corollary*:  $b$  bits of telemetry per decision cap feedback value at  $\Omega\sqrt{(b\ln 2)/2}$ , so compression can preserve but never create feedback value. **(C2)** We show the envelope is *sharp where it matters*: on the canonical binary chain it matches the exact law to  $O(r^3)$  (Proposition 4) — and the battery of §7 *observes* this tightness on real data, including the predicted  $O(r^2)$  correction. **(C3)** We prove no universal equality  $\Delta = f(I)$  exists for bounded objectives (Proposition 5): information is a scalar, value is a geometry. This sharpens against the classical exception — for *logarithmic* (unbounded) utility, Kelly and Barron–Cover [5, 6] show the value of side information *equals*  $I$ ; boundedness is exactly what breaks equality down to an envelope. **(C4)** We give a complete exactness theory for binary state: for *any* bounded objective the value of a delayed sample obeys a *hinge law*  $\Delta(D) = \sum_k \Delta_{s_k}(|C(D)| - C_k^*)^+$  in the lag- $D$  covariance — linear beyond a dead zone, identically zero inside it — recovering the companion’s  $\frac{1}{2}r(D)$  as the symmetric, zero-threshold case and lifting to multi-state monotone objectives by the layer-cake decomposition (§6). The dead zone is a qualitative prediction: for asymmetric states, weakly-correlated feedback is *worthless*, not merely weak. **(C5)** We order the three envelopes (mixing, information, bit-budget) below the master total-variation envelope and prove mixing and information are incomparable — the *preview process* shows  $\beta(D)$  can stay at  $\frac{1}{2}$  forever while the information envelope correctly decays (§5). **(C6)** A nine-series battery across four physical domains confirms the law where surrogates say it should hold, observes the envelope’s tightness, and — read backwards — uses the theorem as a *certified witness* of history information on non-Markov series:  $I(X_t; \text{history}) \geq 2\hat{\Delta}^2$  (§7).

Our techniques are deliberately elementary — Pinsker’s inequality, Jensen, piecewise-linear convex analysis — and we do not claim depth for them. The contribution is the assembly: a single decision-theoretic object explaining an exact systems law, envelopes ordered by what they require (model, samples, or a bit rate), sharpness results tying envelope to law, impossibility results delimiting both, and a falsifiable empirical program run on real processes in five domains.

## 2 Setting and the exact object

Let  $X$  be the current exogenous state taking values in a finite set  $\mathcal{X}$ , with marginal  $\bar{\mu}$ . A controller observes a delayed quantity  $Y$  (a delayed sample  $X_{t-D}$ , a delayed history, or any  $\sigma(\mathcal{F}_{\leq t-D})$ -measurable statistic), chooses an action  $w$ , and receives a bounded one-step reward  $R(w, x)$  with oscillation  $\Omega = \max_{w,x} R - \min_{w,x} R$ . Define the value of a belief  $\mu \in \Delta(\mathcal{X})$ :

$$\psi(\mu) = \max_w \sum_x R(w, x) \mu(x),$$

a pointwise maximum of linear functionals, hence convex; and the posterior  $\mu_Y = \mathbb{P}(X \in \cdot \mid Y)$ , with  $\mathbb{E}[\mu_Y] = \bar{\mu}$  by the tower property. The optimal  $Y$ -measurable controller attains  $V_Y^* = \mathbb{E}[\psi(\mu_Y)]$ ; the

best static controller attains  $V_{\text{stat}} = \psi(\bar{\mu})$ . The exact object of this paper is the **Bayes envelope**

$$\Delta_Y = V_Y^* - V_{\text{stat}} = \mathbb{E}[\psi(\mu_Y)] - \psi(\bar{\mu}) \geq 0, \quad (1)$$

nonnegative by Jensen. Everything that follows is an envelope on (1); the companion paper's exact law is the case where (1) admits a closed form.

**Lemma 1** (Lipschitz value [1, App. A]).  $|\psi(\mu) - \psi(\mu')| \leq \Omega \|\mu - \mu'\|_{\text{TV}}$ , hence  $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$ .

The total-variation route specializes to the  $\beta$ -mixing converse of [1] when  $Y$  is the full delayed history:  $\Delta(D) \leq \Omega \beta(D)$ .

### 3 The predictive-information envelope

**Theorem 2** (Predictive-information bound). *For any bounded objective with oscillation  $\Omega$  and any delayed observable  $Y$ ,*

$$0 \leq \Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Y)} \quad (I \text{ in nats; in bits, } \Omega \sqrt{\frac{\ln 2}{2} I_{\text{bits}}}). \quad (2)$$

*Proof.* By Lemma 1,  $\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$ . Pinsker's inequality gives, for every realization  $y$ ,  $\|\mu_y - \bar{\mu}\|_{\text{TV}} \leq \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_y \|\bar{\mu})}$ . Taking the expectation over  $Y$  and applying Jensen to the concave square root,

$$\mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}} \leq \mathbb{E} \sqrt{\frac{1}{2} D_{\text{KL}}(\mu_Y \|\bar{\mu})} \leq \sqrt{\frac{1}{2} \mathbb{E} D_{\text{KL}}(\mu_Y \|\bar{\mu})} = \sqrt{\frac{1}{2} I(X; Y)},$$

the last equality being the definition of mutual information as the expected KL divergence of posterior from prior.  $\square$

**Corollary 3** (Bit-limited telemetry). *If the sensing path delivers at most  $b$  bits per decision —  $Y = g(Z)$  with  $Z$  the raw delayed signal and  $H(Y) \leq b \ln 2$  nats — then  $\Delta_Y \leq \Omega \sqrt{(b \ln 2)/2}$ , and moreover  $\Delta_Y \leq \Omega \sqrt{\frac{1}{2} I(X; Z)}$  regardless of  $g$  (data processing). Compression can preserve feedback value; no encoding can create it.*

**Proposition 4** (Leading-order tightness in the canonical case). *For the symmetric two-state Markov chain with lag- $D$  autocorrelation  $r = r(D)$ ,*

$$I(X_t; X_{t-D}) = \frac{1}{2} [(1+r) \ln(1+r) + (1-r) \ln(1-r)] = \frac{r^2}{2} + \frac{r^4}{12} + O(r^6),$$

so the right side of (2) equals  $\Omega(\frac{r}{2} + O(r^3))$ , matching the exact law  $\Delta(D) = \frac{1}{2} r(D)$  of [1] to leading order (at the objective normalization  $\Omega = 1$ ). The information envelope is not loose where it matters.

*Proof.* The joint distribution places mass  $\frac{1+r}{4}$  on each agreeing pair and  $\frac{1-r}{4}$  on each disagreeing pair; the stated  $I$  follows by direct computation, and the expansion from  $(1 \pm r) \ln(1 \pm r) = \pm r + \frac{r^2}{2} \mp \frac{r^3}{6} + \dots$ . Then  $\sqrt{\frac{1}{2} I} = \frac{r}{2} \sqrt{1 + \frac{r^2}{6} + O(r^4)}$ .  $\square$

## 4 No universal equality $\Delta = f(I)$

**Proposition 5** (Information is a scalar; value is a geometry). *There is a state space, a bounded objective, and two observations  $Y_1, Y_2$  with  $I(X; Y_1) = I(X; Y_2) > 0$  but  $\Delta_{Y_1} = \frac{\Omega}{2} \cdot \frac{1}{2} > 0 = \Delta_{Y_2}$ . Hence no function  $f$  with  $\Delta_Y = f(I(X; Y))$  exists for all experiments.*

*Proof.* Let  $X = (B_1, B_2)$  be two independent fair bits and let the objective depend only on  $B_1$ : actions  $w \in \{0, 1\}$ ,  $R(w, x) = \Omega \mathbf{1}[w = B_1]$  (so the oscillation is  $\Omega$ ). Static value:  $\Omega/2$ . Let  $Y_1 = B_1$  and  $Y_2 = B_2$ . Both have  $I(X; Y_i) = \ln 2$ . Observing  $Y_1$  yields  $V^* = \Omega$ , so  $\Delta_{Y_1} = \Omega/2$ ; observing  $Y_2$  leaves the posterior over  $B_1$  uniform, so  $\Delta_{Y_2} = 0$ . The two experiments share every information quantity in  $I(X; Y)$  yet differ maximally in value: the envelope (2) is attained within a constant on one and vacuous on the other.  $\square$

**Remark 6.** *This is the precise sense in which the companion paper’s binary law is special: there the state is one decision-relevant bit, so the geometry collapses to a one-parameter family and the envelope inverts to an exact closed form. The general exact object remains (1).*

## 5 Envelope comparison

The three envelopes are not interchangeable, and their relationship is an ordered pair of relaxations of one master quantity.

**Proposition 7** (Master envelope and universal ordering). *For any delayed observable  $Y$  measurable w.r.t. the delayed history  $\mathcal{F}_{\leq t-D}$ ,*

$$\Delta_Y \leq \Omega \mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}} \leq \Omega \min \left\{ \beta(D), \sqrt{\frac{1}{2} I(X; Y)} \right\},$$

where  $\beta(D)$  is the  $\beta$ -mixing coefficient of the state process at lag  $D$ . The expected posterior movement  $\mathbb{E} \|\mu_Y - \bar{\mu}\|_{\text{TV}}$  is the master envelope; the mixing and information envelopes are its two relaxations.

*Proof.* The first inequality is Lemma 1. For the second: since  $Y$  is  $\mathcal{F}_{\leq t-D}$ -measurable and  $\sigma(X) \subseteq \mathcal{F}_{\geq t}$ , conditioning on the coarser  $Y$  can only shrink expected posterior movement, and  $\mathbb{E} \|\mu_{\mathcal{F}_{\leq t-D}} - \bar{\mu}\|_{\text{TV}} \leq \beta(D)$  because the  $\beta$ -coefficient takes its supremum over the larger future field [1, App. A]. The information relaxation is the Pinsker step in the proof of Theorem 2.  $\square$

**Proposition 8** (The two relaxations are incomparable). *Neither relaxation dominates the other.*

- (i) *Mixing strictly tighter: for the symmetric two-state chain with  $Y = X_{t-D}$ ,  $\beta(D) = \frac{1}{2}r$  while  $\sqrt{\frac{1}{2}I} = \frac{r}{2}\sqrt{1 + \frac{r^2}{6}} + O(r^4) > \beta(D)$  for every  $r \in (0, 1)$ .*
- (ii) *Information arbitrarily tighter: for every  $K \geq 2$  there is a stationary state process and objective with  $\beta(D) = \frac{1}{2}$  for all  $D$  (the mixing envelope never decays) while  $\sqrt{\frac{1}{2}I(X; \mathcal{F}_{\leq t-D})} = \sqrt{\ln 2 / (2K)} \rightarrow 0$ , and indeed  $\Delta = \Omega / (2K)$ .*

*Proof.* (i) is Proposition 4 together with  $\beta(D) = \frac{1}{2}r(D)$  for the binary chain [1, Cor. binary]. (ii) *The preview process.* Let  $c_t$  be i.i.d. fair bits, let  $\Theta \sim \text{Unif}\{0, \dots, K-1\}$  be an independent phase, and let the observable state be  $X_t = (c_t, p_t)$  with

$$p_t = \begin{cases} c_{t+m} & t \equiv \Theta \pmod{K} \\ \perp & \text{otherwise,} \end{cases} \quad m = D + K,$$

a sensor that occasionally publishes a *forecast*  $m$  steps ahead. The objective depends only on the current payoff bit:  $R(w, x) = \Omega \mathbf{1}[w = c]$ . *Mixing*: the window  $[t - m, t - D]$  has length  $K$ , hence contains exactly one preview slot  $s \equiv \Theta$ ; that preview reveals  $c_{s+m} \in \{c_t, \dots, c_{t+K}\}$ , a present-or-future payoff bit, so some event in  $\mathcal{F}_{\geq t}$  has conditional probability in  $\{0, 1\}$  against marginal  $\frac{1}{2}$ :  $\beta(D) = \frac{1}{2}$  for every  $D$ . *Information*: the delayed history determines  $c_t$  exactly when  $t - m \equiv \Theta \pmod{K}$  (probability  $1/K$  over the phase) and is independent of  $c_t$  otherwise, so  $I(c_t; \mathcal{F}_{\leq t-D}) = \frac{\ln 2}{K}$ . *Value*: feedback wins  $\Omega/2$  on the revealed fraction  $1/K$  and nothing elsewhere:  $\Delta = \Omega/(2K)$ . The mixing envelope is vacuous; the information envelope  $\Omega \sqrt{\ln 2/(2K)}$  decays with  $K$  (at rate  $K^{-1/2}$  against the true  $K^{-1}$  — loose, but not blind).  $\square$

**Remark 9** (Why keep the information form at all). *Proposition 7 shows the TV master envelope is always at least as tight, and (i) shows mixing can beat information. Three properties are the information envelope's own: it is estimable — the plug-in  $\hat{I}(X_t; X_{t-D})$  in §7 needs only pair counts, while the  $\beta$ -coefficient's supremum over the future  $\sigma$ -field is not estimable without a process model; it composes with channels — data processing gives Corollary 3, and no analogous statement exists for  $\beta$  (mixing has no channel semantics); and it adds across sensors via the chain rule, giving budget statements for multi-sensor telemetry. In short: mixing is the sharper analysis tool when a model is in hand; information is the operational one when only data or a bit budget is.*

| Envelope    | Bound on $\Delta_Y$                                   | Needs             | Blind spot                             |
|-------------|-------------------------------------------------------|-------------------|----------------------------------------|
| master (TV) | $\Omega \mathbb{E} \ \mu_Y - \bar{\mu}\ _{\text{TV}}$ | posterior process | not estimable directly                 |
| mixing      | $\Omega \beta(D)$                                     | process model     | future events beyond $X_t$ (Prop. 8ii) |
| information | $\Omega \sqrt{I(X; Y)/2}$                             | pair samples      | decision geometry (Prop. 5)            |
| bit budget  | $\Omega \sqrt{(b \ln 2)/2}$                           | channel rate only | everything but the rate                |

Table 1: The envelope family. Each successive row needs less knowledge and pays for it in tightness; the battery of §7 measures the first three on real processes.

## 6 Exactness: a complete account for binary state, and its limits

When does the Bayes envelope admit a closed form — an exact *law* rather than an envelope? For binary state the answer is: always, and the form is a hinge.

**Theorem 10** (The hinge law). *Let  $X_t \in \{0, 1\}$  be stationary with  $\pi = \mathbb{P}(X_t=1) \in (0, 1)$ , let  $Y = X_{t-D}$ , and write  $C = C(D) = \text{Cov}(X_t, X_{t-D})$ . For any bounded objective, the value function  $\psi$  is convex piecewise-linear in the scalar belief  $p = \mathbb{P}(X_t=1 \mid \cdot)$ , with kinks  $p_1^* < \dots < p_K^*$  and slope jumps  $\Delta s_k > 0$ . Then, exactly,*

$$\Delta(D) = \sum_{k=1}^K \Delta s_k \left( |C(D)| - C_k^* \right)^+, \quad C_k^* = \begin{cases} (p_k^* - \pi) \pi & \pi \leq p_k^*, C \geq 0 \\ (\pi - p_k^*)(1 - \pi) & \pi > p_k^*, C \geq 0, \end{cases}$$

with the mirrored thresholds for  $C < 0$ . In particular  $\Delta(D)$  is a nondecreasing piecewise-linear function of  $|C(D)|$  that is **identically zero on the dead zone**  $|C(D)| \leq \min_k C_k^*$ .

*Proof.*  $\psi(p) = \max_w [pR(w, 1) + (1 - p)R(w, 0)]$  is a maximum of finitely many affine functions of  $p$ , hence convex piecewise-linear; write  $\psi(p) = \psi(p_1^*) + s_0(p - p_1^*) + \sum_k \Delta s_k (p - p_k^*)^+$ . Observing  $Y$  produces the two-point posterior  $p_1 = \pi + C/\pi$  (w.p.  $\pi$ ) and  $p_0 = \pi - C/(1 - \pi)$  (w.p.  $1 - \pi$ ).

$\pi$ ), whose mean is  $\pi$ ; the affine part of  $\psi$  therefore cancels in  $\Delta = \mathbb{E}[\psi(p_Y)] - \psi(\pi)$ , leaving  $\Delta = \sum_k \Delta s_k \{\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*)^+\}$ . Fix  $k$  and  $C \geq 0$  (so  $p_0 \leq \pi \leq p_1$ ). If  $\pi \leq p_k^*$ :  $\mathbb{E}(p_Y - p_k^*)^+ = \pi(p_1 - p_k^*)^+ = (C - (p_k^* - \pi)\pi)^+$  and  $(\pi - p_k^*)^+ = 0$ . If  $\pi > p_k^*$ : by the pointwise identity  $(x)^+ - x = (-x)^+$ ,  $\mathbb{E}(p_Y - p_k^*)^+ - (\pi - p_k^*)^+ = \mathbb{E}(p_k^* - p_Y)^+$ , and only the lower posterior can fall below the kink, so this equals  $(1 - \pi)(p_k^* - p_0)^+ = (C - (\pi - p_k^*)(1 - \pi))^+$ . The case  $C < 0$  mirrors with  $p_0$  and  $p_1$  exchanged. Summing over  $k$  gives the statement.  $\square$

**Corollary 11** (The companion law, and why it is threshold-free). *For the symmetric chain ( $\pi = \frac{1}{2}$ ) with the 0/1 prediction objective ( $K=1$ ,  $p^* = \frac{1}{2}$ ,  $\Delta s = 2$ ):  $C^* = 0$  and  $C(D) = \frac{1}{4}r(D)$ , so  $\Delta(D) = 2 \cdot \frac{1}{4}r(D) = \frac{1}{2}r(D)$  — the exact law of [1]. Symmetry places the prior on the kink, which is the unique configuration with no dead zone; the linear-in- $r$  law is the degenerate hinge.*

**Remark 12** (The dead zone is a prediction, not a curiosity). *For  $\pi \neq p^*$  the hinge says weakly-correlated delayed feedback is worthless — not slightly valuable — until  $|C(D)|$  crosses  $C^*$ : below threshold, both posteriors fall on one side of the decision kink and the optimal action never changes. Operationally: a rare-event admission controller ( $\pi$  small,  $p^*$  moderate) gains nothing from telemetry until its sensing age satisfies  $|C(D)| > (p^* - \pi)\pi$  — an exact, testable cutoff for when to stop paying for freshness at all. None of the envelopes of §5 see this: the dead zone is a fact about geometry (posteriors vs. kink), which is precisely what scalar information summaries discard (Proposition 5).*

**Remark 13** (Beyond two states: the layer-cake lift, and the boundary). *The companion’s layer-cake result [1, Prop. 2] lifts binary exactness to capacities  $a_t \in \{0, \dots, C\}$  with monotone threshold objectives: the objective decomposes over layer indicators  $x_t^u = \mathbf{1}[a_t \geq u]$  and the total advantage is the exact sum  $\Delta(D) = \sum_u \pi_u r_u(D)$  — each layer a binary experiment, each governed by its hinge (there with symmetric thresholds, so zero dead zones). The boundary of exactness is now visible from both sides: exact laws hold precisely when the posterior family is effectively one-dimensional (a scalar belief per layer, moved along a line by the observation); the two-bit construction of Proposition 5 is the minimal experiment whose posterior geometry is genuinely two-dimensional, and there the value ceases to be a function of any scalar summary of dependence — correlation or information alike.*

## 7 The law and its envelopes across domains

The companion paper had room for one paragraph of this study; the full protocol and results are reported here. The question has two halves. First, *the law*: is  $\Delta(D) = \frac{1}{2}r(D)$  — exact for symmetric binary Markov capacity — also the operative value of feedback on *real* binary state processes, i.e., does history beyond the single delayed sample add value? Second, *the envelope*: does the measured value respect, and how tightly does it approach, the predictive-information bound of Theorem 2?

### 7.1 Protocol

**States.** Binary regime series from four physical domains plus controls: acoustic source azimuth and voice activity (LOCATA, 120 Hz); sperm-whale coda type (Dominica corpus, per-exchange); GOES space-weather series (X-ray flux, magnetometer, proton flux; 1–5-minute cadence); seismic station regime (ANMO, 1-minute); a synthetic two-state Markov chain (flip probability 0.02) as positive reference; an i.i.d. fair-bit series as null. Each series is binarized by the domain’s registered thresholding (the companion study’s preparation scripts, unchanged).



**The two estimands.** For each lag  $D$  on a per-domain grid: (i) the pooled lag- $D$  autocorrelation  $r(D)$ , giving the law value  $\frac{1}{2}r(D)$ ; (ii) the value of the best *delayed-history* policy  $\hat{\Delta}_{\text{hist}}(D)$ : a  $k=6$ -lag lookup predictor trained and evaluated by blocked cross-validation (5 folds, contiguous blocks, no leakage across fold boundaries), reported as accuracy above chance — the operational value of feedback for the 0/1 prediction objective ( $\Omega = 1$ , for which the binary law reads exactly  $\frac{1}{2}r$ ).

**Surrogate calibration.** The null for “history adds nothing beyond the law” is a population of 60 Markov surrogates per domain, each matched to the real series’ lag-1 transition probabilities and evaluated by the identical pipeline. The gain  $\hat{\Delta}_{\text{hist}} - \frac{1}{2}r$  is called significant only if its maximum over the lag grid exceeds the surrogate maxima at  $p < 0.05$  *and* clears a practical floor of 0.03 (twice the i.i.d. null’s estimator fluctuation) — the dual criterion of the companion study.

**The envelope measurement.** New here: for each domain and lag, the plug-in mutual information  $\hat{I}(X_t; X_{t-D})$  from pooled pair counts ( $2 \times 2$  table; plug-in bias  $\leq 3/(2N)$  nats at the smallest pooled  $N$  in the battery, negligible at the reported scale), and the envelope  $\sqrt{\hat{I}/2}$  of Theorem 2. Because  $\hat{\Delta}_{\text{hist}}$  is a noisy estimate, the registered check is

$$\hat{\Delta}_{\text{hist}}(D) \leq \sqrt{\hat{I}(D)/2} + \varepsilon_{\text{est}},$$

with  $\varepsilon_{\text{est}}$  the maximum absolute fluctuation of the same estimator on the i.i.d. null — the null calibrates the noise floor, so an apparent “violation” smaller than what pure noise produces on a zero-information series is reported as estimator noise, not physics.

## 7.2 Results

Table 2 reports, per domain: the maximum history gain beyond the law with its surrogate  $p$ -value; the per-domain noise bound  $\varepsilon_{\text{dom}}$  (the maximum of the surrogates’ 97.5% gain band — surrogates have zero true history gain by construction and share the domain’s series lengths, so they calibrate its estimator noise); the maximum excess of the history policy over the *single-lag* envelope  $\sqrt{\hat{I}(X_t; X_{t-D})/2}$ ; and the tightness ratio envelope/law (median over lags with  $r > 0.05$ ).

| Domain                 | gain <sub>max</sub> | $p$ | $\varepsilon_{\text{dom}}$ | $\max(\hat{\Delta} - \text{env}_1)$ | env/law | verdict            |
|------------------------|---------------------|-----|----------------------------|-------------------------------------|---------|--------------------|
| Two-state Markov (ref) | 0.030               | .38 | 0.069                      | +0.023                              | 1.001   | law exact          |
| i.i.d. null (control)  | 0.012               | .68 | 0.016                      | +0.012                              | —       | $\Delta \approx 0$ |
| LOCATA azimuth         | 0.008               | .05 | 0.007                      | −0.013                              | 1.099   | law exact          |
| Sperm-whale codas      | 0.046               | .98 | 0.112                      | +0.045                              | 1.010   | law exact          |
| GOES protons           | 0.029               | .77 | 0.048                      | +0.030                              | 0.998   | law exact          |
| LOCATA VAD             | 0.100               | .00 | 0.059                      | <b>+0.128</b>                       | 0.940   | history adds       |
| GOES X-ray flux        | 0.185               | .00 | 0.100                      | +0.071                              | 0.937   | history adds       |
| GOES magnetometer      | 0.147               | .00 | 0.070                      | +0.041                              | 0.972   | history adds       |
| Seismic ANMO           | 0.367               | .00 | 0.037                      | <b>+0.157</b>                       | 0.999   | history adds       |

Table 2: Battery v3 (nine series, four physical domains + controls; `battery_v3.json`). Upper block: the law is the whole value of feedback and the single-lag envelope is respected within noise. Lower block: the surrogate test certifies history value beyond the law ( $p < .05$  and gain above the 0.03 floor), and exactly there the history policy can exceed the *single-sample* envelope — bold where the excess clears  $\varepsilon_{\text{dom}}$ , certifying window information beyond the single lag (see text).

Three readings, in increasing order of consequence.

**The law, confirmed where it should hold.** On the Markov reference and the three domains whose surrogate test finds nothing beyond the law (azimuth, whale codas, protons), the best delayed-history policy sits on  $\frac{1}{2}r(D)$  across the lag grid — the single delayed sample is the whole value of feedback, in acoustics, bioacoustics, and space weather alike.

**Tightness, observed in the wild.** The envelope/law column is Proposition 4 measured: 1.001 on the reference chain (small  $r$ : the envelope sits *on* the law), rising to 1.099 on azimuth exactly as the  $\sqrt{1+r^2/6}$  correction predicts for its slowly-decaying (large- $r$ ) regime. The information envelope is not a loose theoretical courtesy; on Markov-regime data it *is* the law, to within a predicted  $O(r^2)$  factor.

**The envelope as a non-Markovianity witness.** In the lower block the history policy exceeds the single-lag envelope — decisively on VAD (+0.128 against a noise bound of 0.059) and seismic regime (+0.157 vs. 0.037). This is not a counterexample to Theorem 2: the history policy’s observable is the 6-lag window  $Y$ , and the theorem bounds it by  $\sqrt{I(X_t; Y)}/2$  for *that*  $Y$ . Read backwards, the theorem converts the measured exceedance into a *certified information lower bound*:

$$I(X_t; \text{window}) \geq 2 \hat{\Delta}_{\text{hist}}^2,$$

which on the seismic series certifies  $\geq 0.072$  nats of window information at a lag where the single sample carries 0.002 — a  $30\times$  concentration of predictive information in history beyond the last sample (aftershock clustering, in information units). The two blocks of Table 2 are therefore not “pass” and “fail”: they are the theorem operating in both directions — as an envelope on Markov-regime processes, and as a quantitative witness of history structure on non-Markov ones. The dual use is registered in the companion study’s terms: the same surrogate machinery that certifies the law’s exactness upper-block certifies the window-information excess lower-block.

## 8 Related work

*Value of information.* The Bayes envelope is the classical object of statistical decision theory [2, 4]; De Lara and Gossner [3] develop the payoffs–beliefs duality our Lipschitz lemma instantiates. The information envelope itself is assembled from textbook inequalities (Pinsker, Jensen), and we make no claim of depth for the inequality in isolation — kindred moves appear wherever bounded value must be controlled by divergence, e.g. the information-ratio regret analyses of Russo and Van Roy [11]. What we have not found elsewhere is the assembly this paper is about: the envelope *anchored to an exact law* (sharp to  $O(r^3)$ , with the  $O(r^2)$  correction observed on real data), its ordering against mixing envelopes with both separations constructed, and its inversion into a certified information witness.

*The logarithmic exception.* For log-utility gambling the value of side information *equals* mutual information: Kelly [5] for the growth rate, Barron and Cover [6] for the general bound  $\Delta \leq I$  with equality characterizations. Log utility is unbounded, escaping our  $\Omega$ -bounded class; the pair of results brackets the landscape — unbounded log utility buys equality, bounded utility only an envelope, and Proposition 5 shows the gap is essential, not technical.

*Age of information.* AoI [7] treats freshness as the object, with sampling-policy results such as [8] optimizing age processes. Our thesis is complementary: age has decision value only through predictive information (the battery’s i.i.d. null has perfect freshness semantics and zero value), and the hinge law shows even the map from correlation to value can have a dead zone freshness metrics cannot see.

*Delayed information in control.* Delayed-sharing information structures go back to Witsenhausen [9] and are systematized by the common-information approach [10]; remote estimation



studies the estimation face of the same trade [8]. Our setting is the tractable exogenous corner of that world — the action does not affect the state — which is exactly what makes closed forms and universal envelopes available.

*Predictive information.* Bialek, Nemenman and Tishby [12] study  $I(\text{past}; \text{future})$  as a measure of process complexity. Theorems 2 and 10 are, in that vocabulary, statements about what predictive information can and cannot buy a bounded-utility decision maker — with the battery’s witness bound  $I \geq 2\hat{\Delta}^2$  operating as a decision-theoretic estimator of predictive information from policy value alone.

## 9 Conclusion

The companion paper measured one exact law on one system. This paper locates that law inside a small, complete theory: value of delayed information is the Bayes envelope; the envelope is bounded by predictive information (universally), by mixing (given a model), and by the telemetry bit rate (given nothing); it is exactly a hinge in covariance for binary state, exactly a layered sum for monotone multi-state objectives, and exactly nothing — inside the dead zone — for asymmetric weakly-coupled sensing. Beyond one effective dimension, no scalar summary of dependence determines value at all. Each claim is falsifiable, and the nine-series battery exercises all of them: the law holds where surrogates say it must, the envelope’s predicted tightness appears with its  $O(r^2)$  correction, and its violations, where the theory requires them, certify history information in nats. The instrument reads in both directions; that, more than any single inequality, is what we hope survives.

## References

- [1] A. H. Bond. Polite bursting: the value of delayed feedback in age-limited admission control. Under submission, IEEE INFOCOM 2027.
- [2] R. A. Howard. Information value theory. *IEEE Trans. Systems Science and Cybernetics* 2(1):22–26, 1966.
- [3] M. De Lara, O. Gossner. Payoffs-beliefs duality and the value of information. *SIAM J. Optimization* 30(1):287–310, 2020.
- [4] H. Raiffa, R. Schlaifer. *Applied Statistical Decision Theory*. Harvard University, 1961.
- [5] J. L. Kelly, Jr. A new interpretation of information rate. *Bell System Technical Journal* 35:917–926, 1956.
- [6] A. R. Barron, T. M. Cover. A bound on the financial value of information. *IEEE Trans. Information Theory* 34(5):1097–1100, 1988.
- [7] R. D. Yates, Y. Sun, D. R. Brown III, S. K. Kaul, E. Modiano, S. Ulukus. Age of information: an introduction and survey. *IEEE J. Selected Areas in Communications* 39(5):1183–1210, 2021.
- [8] Y. Sun, E. Uysal-Biyikoglu, R. D. Yates, C. E. Koksall, N. B. Shroff. Update or wait: how to keep your data fresh. *IEEE Trans. Information Theory* 63(11):7492–7508, 2017.
- [9] H. S. Witsenhausen. A counterexample in stochastic optimum control. *SIAM J. Control* 6(1):131–147, 1968.

- [10] A. Nayyar, A. Mahajan, D. Teneketzis. Decentralized stochastic control with partial history sharing: a common information approach. *IEEE Trans. Automatic Control* 58(7):1644–1658, 2013.
- [11] D. Russo, B. Van Roy. An information-theoretic analysis of Thompson sampling. *J. Machine Learning Research* 17(68):1–30, 2016.
- [12] W. Bialek, I. Nemenman, N. Tishby. Predictability, complexity, and learning. *Neural Computation* 13:2409–2463, 2001.